KCE PATHWAYS · GRADES 4–5 · UNIT 3: BUILDING THINGS · WEEK 11 · MATH

The Dairy Farm Schedule — Teacher Lesson Plan & Answer Key

Learning objectives: Students find factors and factor pairs, identify prime and composite numbers (1 is neither) using divisibility rules for 2, 3, 5, 9, and 10, and find common factors (GCF) and common multiples (LCM) to solve one- and two-step word problems. Students also explain their reasoning, find mistakes in others' work, and correct punctuation, capitalization, and grammar in math sentences.

Duration: 35–45 minutes
Materials: Computer or tablet with the lesson open · pencil · printed Pre & Post sheet

Lesson flow (as it appears on screen)

1. Slide 0 — Intro and name: meet Nadia on her family's Wisconsin dairy farm; student types first and last name so results are sent.
2. Slide 1 — Warm-Up: Unit 2 Review (12 pts): standard algorithm, long division with 2-digit divisors, unlike-denominator fractions, decimals. If a student misses two or more, pause and rework one on paper before moving on.
3. Slide 2 — Pencil Break (Pre): student completes the Pre-Lesson Warm-Up sheet off screen.
4. Slide 3 — Hook: why a dairy farm runs on a schedule (cows milked at least twice a day); Nadia's to-do list previews every problem type.
5. Slide 4 — Picture It: arrays of 36 calves show the factor pairs of 36. Have the student say each pair aloud and notice the pairs stop repeating at 6 × 6.
6. Slide 5 — Practice: factors and factor pairs (12 pts). Encourage listing factor pairs in order from 1 up.
7. Slide 6 — Teach: prime vs. composite, why 1 is neither, 2 is the only even prime, divisibility rules for 2, 3, 5, 9, 10, and the 51 trap (51 = 3 × 17).
8. Slide 7 — Practice: prime/composite, divisibility, and a find-the-mistake item (12 pts).
9. Slide 8 — Vocabulary flip cards: factor, factor pair, multiple, prime, composite, divisible, GCF, LCM.
10. Slide 9 — Vocabulary check (6 pts).
11. Slide 10 — Teach: multiples and the LCM using the feed truck (every 4 days) and vet (every 6 days) calendar; GCF using 24 curd bags and 36 milk bottles. Key cue: "together on the same day" means LCM; "equal groups, nothing left over" means GCF.
12. Slide 11 — Practice: LCM and GCF (12 pts).
13. Slide 12 — Real World: hay stacks / calf pens, two-step factor problem (5 pts).
14. Slide 13 — Real World: packing identical crates with the GCF (5 pts).
15. Slide 14 — Think It Through: "which explanation is correct?" and "find the mistake" (10 pts). Ask the student to say why each wrong choice is wrong.
16. Slide 15 — Fix the Sentence (8 pts): four sentences on quotation marks in dialogue, its vs. it's, commas in a series, and subject-verb agreement. The answer must match exactly (straight or curly quotes both work).
17. Slide 16 — Match-Up Round (10 pts): 5 clue-to-answer matches, 2 pts each on the first try.
18. Slide 17 — Speed Trail (bonus XP only).
19. Slide 18 — Looking Back reflection (bonus XP only).
20. Slide 19 — Boss Battle (8 pts): two-step LCM and GCF problems on a delivery day.

21. Slide 20 — Pencil Break (Post): student completes the Post-Lesson Challenge sheet off screen.
22. Slide 21 — Results: score out of 100 sent to KingCretot Experience; 85+ shows the farm-planner congratulations with confetti; below 85 shows encouragement, and Play Again gives the next question set.

Lesson answer key with reasoning

Every scored item the student sees, in order, across all three rotating question sets. The "why" is the same reasoning shown on screen after a second missed try.

WARM-UP: UNIT 2 REVIEW (3 POINTS EACH)

ID	Question	Answer	Why (explain to student)
r1_s0	Set 1: Standard algorithm (Week 6): type the product. 34 × 26 =	**884**	34 × 6 = 204 and 34 × 20 = 680. Add the partial products: 204 + 680 = 884.
r2_s0	Set 1: Long division (Week 7): 672 ÷ 24 = ?	**28**	24 goes into 67 two times (48), leaving 19. Bring down the 2 to make 192. 24 x 8 = 192. So the quotient is 28.
r3_s0	Set 1: Unlike denominators (Week 8): 1/2 + 1/3 = ?	**5/6**	Use the common denominator 6: 1/2 = 3/6 and 1/3 = 2/6. Then 3/6 + 2/6 = 5/6.
r4_s0	Set 1: Decimals (Week 9): type the product. 0.4 × 6 =	**2.4**	4 × 6 = 24. 0.4 has 1 decimal place, so the answer is 2.4.
r1_s1	Set 2: Standard algorithm (Week 6): type the product. 43 × 25 =	**1,075**	43 × 5 = 215 and 43 × 20 = 860. Add the partial products: 215 + 860 = 1,075.
r2_s1	Set 2: Long division (Week 7): 864 ÷ 32 = ?	**27**	32 goes into 86 two times (64), leaving 22. Bring down the 4 to make 224. 32 x 7 = 224. So the quotient is 27.
r3_s1	Set 2: Unlike denominators (Week 8): 1/4 + 2/3 = ?	**11/12**	Use the common denominator 12: 1/4 = 3/12 and 2/3 = 8/12. Then 3/12 + 8/12 = 11/12.
r4_s1	Set 2: Fractions to decimals (Week 9): type 3/4 as a decimal. 3/4 =	**0.75**	3/4 means 3 divided by 4, which is 0.75.
r1_s2	Set 3: Standard algorithm (Week 6): type the product. 52 × 37 =	**1,924**	52 × 7 = 364 and 52 × 30 = 1,560. Add the partial products: 364 + 1,560 = 1,924.
r2_s2	Set 3: Long division (Week 7): 735 ÷ 21 = ?	**35**	21 goes into 73 three times (63), leaving 10. Bring down the 5 to make 105. 21 × 5 = 105. So the quotient is 35.
r3_s2	Set 3: Unlike denominators (Week 8): 2/3 − 1/2 = ?	**1/6**	Use the common denominator 6: 2/3 = 4/6 and 1/2 = 3/6. Then 4/6 - 3/6 = 1/6.
r4_s2	Set 3: Decimals (Week 9): type the product. 1.5 × 4 =	**6**	15 × 4 = 60. 1.5 has 1 decimal place, so 6.0, which is 6.

YOUR TURN: FACTORS AND FACTOR PAIRS (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
f1_s0	Set 1: Which of these is NOT a factor pair of 36?	**5 × 7**	5 × 7 = 35, not 36. The others all multiply to 36.
f2_s0	Set 1: How many factors does 24 have? Type the number. Number of factors of 24:	**8**	Factor pairs of 24: 1 × 24, 2 × 12, 3 × 8, 4 × 6. That gives 8 factors: 1, 2, 3, 4, 6, 8, 12, 24.

ID	Question	Answer	Why (explain to student)
f3_s0	Set 1: Which list shows ALL the factors of 18?	**1, 2, 3, 6, 9, 18**	Factor pairs of 18: 1 × 18, 2 × 9, 3 × 6. So the factors are 1, 2, 3, 6, 9, 18. 4 is not a factor because 18 divided by 4 has a remainder.
f1_s1	Set 2: Which of these is NOT a factor pair of 48?	**7 × 7**	7 × 7 = 49, not 48. The others all multiply to 48.
f2_s1	Set 2: How many factors does 20 have? Type the number. Number of factors of 20:	**6**	Factor pairs of 20: 1 × 20, 2 × 10, 4 × 5. That gives 6 factors: 1, 2, 4, 5, 10, 20.
f3_s1	Set 2: Which list shows ALL the factors of 28?	**1, 2, 4, 7, 14, 28**	Factor pairs of 28: 1 × 28, 2 × 14, 4 × 7. So the factors are 1, 2, 4, 7, 14, 28. 3 is not a factor because 28 divided by 3 has a remainder.
f1_s2	Set 3: Which of these is NOT a factor pair of 40?	**6 × 7**	6 × 7 = 42, not 40. The others all multiply to 40.
f2_s2	Set 3: How many factors does 30 have? Type the number. Number of factors of 30:	**8**	Factor pairs of 30: 1 × 30, 2 × 15, 3 × 10, 5 × 6. That gives 8 factors: 1, 2, 3, 5, 6, 10, 15, 30.
f3_s2	Set 3: Which list shows ALL the factors of 32?	**1, 2, 4, 8, 16, 32**	Factor pairs of 32: 1 × 32, 2 × 16, 4 × 8. So the factors are 1, 2, 4, 8, 16, 32. 6 is not a factor because 32 divided by 6 has a remainder.

YOUR TURN: PRIME OR COMPOSITE? (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
p1_s0	Set 1: Nadia is checking cow ID tags. Which cow's ID number is prime?	**29**	29 has only two factors, 1 and 29. 27 = 3 × 9, 21 = 3 × 7, and 33 = 3 × 11, so those are composite.
p2_s0	Set 1: Use the divisibility rule for 3. Which number is divisible by 3?	**471**	Add the digits: 4 + 7 + 1 = 12, and 12 is divisible by 3, so 471 is too. The other digit sums are 13, 11, and 14, which are not divisible by 3.
p3_s0	Set 1: Find the mistake: a classmate says 51 is prime. Why is that wrong?	**51 = 3 × 17, so it has more than two factors**	The digits of 51 add to 6, so 51 is divisible by 3: 51 = 3 × 17. Its factors are 1, 3, 17, 51, which is more than two, so 51 is composite. (Odd numbers like 7 and 13 CAN be prime.)
p1_s1	Set 2: Nadia is checking cow ID tags. Which cow's ID number is composite?	**39**	39 = 3 × 13, so it has more than two factors. 31, 37, and 41 each have only two factors, so they are prime.
p2_s1	Set 2: Use the divisibility rule for 9. Which number is divisible by 9?	**738**	Add the digits: 7 + 3 + 8 = 18, and 18 is divisible by 9, so 738 is too. The other digit sums are 17, 19, and 19.
p3_s1	Set 2: Find the mistake: Ben says 1 is a prime number. Why is Ben wrong?	**A prime has exactly two factors, but 1 has only one factor**	A prime number has exactly two different factors: 1 and itself. The number 1 has only ONE factor (1), so it is neither prime nor composite.

ID	Question	Answer	Why (explain to student)
p1_s2	Set 3: Nadia is checking cow ID tags. Which cow's ID number is prime?	**47**	47 has only two factors, 1 and 47. 49 = 7 × 7, 45 = 5 × 9, and 51 = 3 × 17, so those are composite.
p2_s2	Set 3: Which number is divisible by 2, by 5, AND by 10?	**340**	A number divisible by 10 ends in 0. 340 ends in 0, so it is divisible by 2, 5, and 10. 345 is not even, and 342 and 354 do not end in 0 or 5.
p3_s2	Set 3: Find the mistake: Maya says 2 cannot be prime because it is even. Why is Maya wrong?	**The only factors of 2 are 1 and 2, so 2 is prime**	2 has exactly two factors, 1 and 2, so it is prime. It is the only even prime number. Every other even number has 2 as an extra factor.

YOUR TURN: USING THE WORDS (3 POINTS EACH)

ID	Question	Answer	Why (explain to student)
v1_s0	Set 1: A whole number with exactly two factors, 1 and itself, is called a ___.	**Prime number**	Exactly two factors (1 and itself) is the definition of a prime number.
v2_s0	Set 1: Type your answer: numbers you multiply together to get a product are called ___.	**factors**	Numbers you multiply to get a product are factors. In 4 × 9 = 36, 4 and 9 are factors of 36.
v1_s1	Set 2: A whole number greater than 1 with MORE than two factors is called ___.	**Composite**	More than two factors means the number is composite.
v2_s1	Set 2: Type your answer: 12, 24, and 36 are all ___ of 12.	**multiples**	You get 12, 24, and 36 by multiplying 12 by 1, 2, and 3, so they are multiples of 12.
v1_s2	Set 3: The smallest number that is a multiple of two numbers is their ___.	**Least common multiple (LCM)**	The least common multiple is the smallest number both numbers divide into evenly.
v2_s2	Set 3: Type your answer: 36 ÷ 4 has no remainder, so 36 is ___ by 4.	**divisible**	When a number divides evenly with no remainder, the first number is divisible by the second.

YOUR TURN: LCM AND GCF (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
m1_s0	Set 1: Type the least common multiple (LCM) of 4 and 6. LCM of 4 and 6 =	**12**	Multiples of 4: 4, 8, 12... Multiples of 6: 6, 12... The first one they share is 12.
m2_s0	Set 1: Type the greatest common factor (GCF) of 24 and 36. GCF of 24 and 36 =	**12**	Factors of 24: 1, 2, 3, 4, 6, 8, 12, 24. Factors of 36: 1, 2, 3, 4, 6, 9, 12, 18, 36. The greatest one they share is 12.
m3_s0	Set 1: Which number is a common multiple of 3 and 5?	**15**	15 = 3 × 5, so it is a multiple of both. 10 and 20 are multiples of 5 but not 3, and 8 is a multiple of neither.
m1_s1	Set 2: Type the least common multiple (LCM) of 6 and 8. LCM of 6 and 8 =	**24**	Multiples of 6: 6, 12, 18, 24... Multiples of 8: 8, 16, 24... The first one they share is 24.
m2_s1	Set 2: Type the greatest common factor (GCF) of 18 and 24. GCF of 18 and 24 =	**6**	Factors of 18: 1, 2, 3, 6, 9, 18. Factors of 24: 1, 2, 3, 4, 6, 8, 12, 24. The greatest one they share is 6.

ID	Question	Answer	Why (explain to student)
m3_s1	Set 2: Which number is a common factor of 16 and 20?	**4**	16 divided by 4 is 4 and 20 divided by 4 is 5, so 4 divides both. 8 does not divide 20, and 5 and 10 do not divide 16.
m1_s2	Set 3: Type the least common multiple (LCM) of 5 and 6. LCM of 5 and 6 =	**30**	Multiples of 5: 5, 10, 15, 20, 25, 30... Multiples of 6: 6, 12, 18, 24, 30... The first one they share is 30.
m2_s2	Set 3: Type the greatest common factor (GCF) of 20 and 30. GCF of 20 and 30 =	**10**	Factors of 20: 1, 2, 4, 5, 10, 20. Factors of 30: 1, 2, 3, 5, 6, 10, 15, 30. The greatest one they share is 10.
m3_s2	Set 3: Which number is a common multiple of 4 and 10?	**20**	20 = 4 × 5 and 20 = 10 × 2, so it is a multiple of both. 30 is not a multiple of 4, 24 is not a multiple of 10, and 14 is neither.

REAL WORLD: STACKS AND PENS (5 POINTS)

ID	Question	Answer	Why (explain to student)
w1_s0	Set 1: Nadia has 48 hay bales to split into equal stacks. Each stack must have more than 6 bales but fewer than 12 bales. Two steps: find the stack size that works, then type how many stacks Nadia makes. Number of stacks:	**6**	Factors of 48: 1, 2, 3, 4, 6, 8, 12, 16, 24, 48. The only one between 6 and 12 is 8, so each stack has 8 bales. 48 divided by 8 = 6 stacks.
w1_s1	Set 2: Nadia is arranging 36 calves into equal rows of pens. She wants more than 3 rows but fewer than 6 rows. Two steps: find the number of rows that works, then type how many calves are in each row. Calves in each row:	**9**	Factors of 36: 1, 2, 3, 4, 6, 9, 12, 18, 36. The only number of rows between 3 and 6 is 4 (5 is not a factor of 36). 36 divided by 4 = 9 calves in each row.
w1_s2	Set 3: Nadia has 72 hay bales to split into equal stacks. Each stack must have more than 15 bales but fewer than 20 bales. Two steps: find the stack size that works, then type how many stacks Nadia makes. Number of stacks:	**4**	Factors of 72: 1, 2, 3, 4, 6, 8, 9, 12, 18, 24, 36, 72. The only one between 15 and 20 is 18, so each stack has 18 bales. 72 divided by 18 = 4 stacks.

REAL WORLD: PACKING THE CRATES (5 POINTS)

ID	Question	Answer	Why (explain to student)
w2_s0	Set 1: Nadia packs 24 bags of cheese curds and 36 milk bottles into identical crates. Every crate must hold the same number of curd bags and the same number of bottles, with nothing left over. What is the GREATEST number of crates Nadia can fill?	**12**	The number of crates must divide both 24 and 36 evenly, so it is a common factor. The greatest common factor of 24 and 36 is 12, so she can fill 12 crates.
w2_s1	Set 2: Nadia packs 18 bags of cheese curds and 30 milk bottles into identical crates. Every crate must hold the same number of curd bags and the same number of bottles, with nothing left over. What is the GREATEST number of crates Nadia can fill?	**6**	The number of crates must divide both 18 and 30 evenly. The greatest common factor of 18 and 30 is 6, so she can fill 6 crates.

ID	Question	Answer	Why (explain to student)
w2_s2	Set 3: Nadia packs 32 bags of cheese curds and 40 milk bottles into identical crates. Every crate must hold the same number of curd bags and the same number of bottles, with nothing left over. What is the GREATEST number of crates Nadia can fill?	**8**	The number of crates must divide both 32 and 40 evenly. The greatest common factor of 32 and 40 is 8, so she can fill 8 crates.

THINK IT THROUGH: EXPLAIN AND FIND THE MISTAKE (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
x1_s0	Set 1: Which explanation is correct? The feed truck comes every 4 days and the vet comes every 6 days. Both came today. When will they next come on the same day?	**Day 12, because 12 is the least common multiple of 4 and 6**	Truck days: 4, 8, 12... Vet days: 6, 12... The first shared day is 12, the LCM. Day 24 is also a shared day, but it is not the NEXT one.
x2_s0	Set 1: Find the mistake: Ravi listed the factors of 36 as 1, 2, 3, 4, 6, 12, 18, 36. What did he do wrong?	**He left out 9, because 4 × 9 = 36**	Check the factor pairs: 1 × 36, 2 × 18, 3 x 12, 4 × 9, 6 × 6. Ravi forgot 9, the partner of 4. (8 is not a factor: 36 divided by 8 has a remainder.)
x1_s1	Set 2: Which explanation is correct? Is 117 divisible by 9?	**Yes, because its digits add to 9, and 9 is divisible by 9**	The rule for 9: add the digits. 1 + 1 + 7 = 9, which is divisible by 9, so 117 is too (117 = 9 × 13). Being odd or the last digit does not decide it.
x2_s1	Set 2: Find the mistake: Lena says the GCF of 12 and 18 is 3. What did she do wrong?	**3 is a common factor, but 6 is a greater one, so the GCF is 6**	Common factors of 12 and 18: 1, 2, 3, 6. Lena found a common factor, but not the GREATEST one. The GCF is 6.
x1_s2	Set 3: Which explanation is correct? Why is 57 composite?	**57 = 3 × 19, so it has factors other than 1 and 57**	The digits of 57 add to 12, so 3 is a factor: 57 = 3 × 19. A number with factors besides 1 and itself is composite.
x2_s2	Set 3: Find the mistake: Omar says the LCM of 6 and 9 is 54 because 6 × 9 = 54. What did he do wrong?	**54 is a common multiple, but 18 is smaller, so the LCM is 18**	Multiples of 6: 6, 12, 18... Multiples of 9: 9, 18... The first one they share is 18. Multiplying the two numbers gives a common multiple, but not always the least one.

FIX THE SENTENCE (2 POINTS EACH)

ID	Question	Answer	Why (explain to student)
s1_s0	Set 1: Fix the sentence: nadia said, 36 has nine factors	**Nadia said, "36 has nine factors."**	Capitalize the name Nadia, put quotation marks around her exact words, and put the period inside the closing quotation mark.
s2_s0	Set 1: Fix the sentence: The number 7 is prime because it's only factors are 1 and 7.	**The number 7 is prime because its only factors are 1 and 7.**	Its (no apostrophe) shows ownership: the factors belong to 7. It's always means it is.
s3_s0	Set 1: Fix the sentence: The factors of 10 are 1 2 5 and 10.	**The factors of 10 are 1, 2, 5, and 10. OR The factors of 10 are 1, 2, 5 and 10.**	Use commas to separate the items in a series: 1, 2, 5, and 10.

ID	Question	Answer	Why (explain to student)
s4_s0	Set 1: Fix the sentence: nadia and her dad counts the factors of 48	**Nadia and her dad count the factors of 48.**	Capitalize the first word and the name Nadia. Two people (Nadia and her dad) need the verb count, not counts. End with a period.
s1_s1	Set 2: Fix the sentence: dad asked, is 51 a prime number	**Dad asked, "Is 51 a prime number?"**	Capitalize the first word, put quotation marks around the exact words, capitalize the first word of the quote, and put the question mark inside the closing quotation mark.
s2_s1	Set 2: Fix the sentence: Its easy to see that 48 is even because it ends in 8.	**It's easy to see that 48 is even because it ends in 8.**	It's means it is: it is easy to see. Use the apostrophe.
s3_s1	Set 2: Fix the sentence: The first three multiples of 6 are 6 12 and 18.	**The first three multiples of 6 are 6, 12, and 18. OR The first three multiples of 6 are 6, 12 and 18.**	Use commas to separate the items in a series: 6, 12, and 18.
s4_s1	Set 2: Fix the sentence: every even number greater than 2 are composite	**Every even number greater than 2 is composite.**	Capitalize the first word. The subject is every even number (singular), so use is, not are. End with a period.
s1_s2	Set 3: Fix the sentence: nadia said, the number 1 is neither prime nor composite	**Nadia said, "The number 1 is neither prime nor composite."**	Capitalize Nadia, put quotation marks around her exact words, capitalize the first word of the quote, and put the period inside the closing quotation mark.
s2_s2	Set 3: Fix the sentence: A number is divisible by 5 when it's last digit is 0 or 5.	**A number is divisible by 5 when its last digit is 0 or 5.**	Its (no apostrophe) shows ownership: the last digit belongs to the number. It's means it is.
s3_s2	Set 3: Fix the sentence: The common factors of 24 and 36 are 1 2 3 4 6 and 12.	**The common factors of 24 and 36 are 1, 2, 3, 4, 6, and 12. OR The common factors of 24 and 36 are 1, 2, 3, 4, 6 and 12.**	Use commas to separate the items in a series: 1, 2, 3, 4, 6, and 12.
s4_s2	Set 3: Fix the sentence: the feed truck and the vet both comes on day 12	**The feed truck and the vet both come on day 12.**	Capitalize the first word. Two subjects (the feed truck and the vet) need the verb come, not comes. End with a period.

MATCH-UP ROUND (2 POINTS PER FIRST-TRY MATCH, 10 TOTAL)

ID	Question	Answer	Why (explain to student)
match1_s0	Set 1: Factors of 9	**1, 3, 9**	1 × 9 and 3 × 3, so the factors are 1, 3, 9.
match2_s0	Set 1: GCF of 8 and 12	**4**	Factors of 8: 1, 2, 4, 8. Factors of 12: 1, 2, 3, 4, 6, 12. Greatest shared: 4.
match3_s0	Set 1: LCM of 4 and 6	**12**	Multiples of 4: 4, 8, 12. Multiples of 6: 6, 12. First shared: 12.

ID	Question	Answer	Why (explain to student)
match4_s0	Set 1: Smallest prime number	**2**	2 has exactly two factors, 1 and 2. 1 is not prime.
match5_s0	Set 1: Next prime after 23	**29**	24, 25, 26, 27, 28 are composite; 29 has only the factors 1 and 29.
match1_s1	Set 2: Factors of 15	**1, 3, 5, 15**	1 × 15 and 3 × 5.
match2_s1	Set 2: GCF of 18 and 24	**6**	Common factors 1, 2, 3, 6. Greatest: 6.
match3_s1	Set 2: LCM of 3 and 5	**15**	Multiples of 3: 3, 6, 9, 12, 15. Multiples of 5: 5, 10, 15. First shared: 15.
match4_s1	Set 2: Neither prime nor composite	**1**	1 has only one factor.
match5_s1	Set 2: A factor pair of 21 (not 1 × 21)	**3 × 7**	3 × 7 = 21.
match1_s2	Set 3: Factors of 16	**1, 2, 4, 8, 16**	1 × 16, 2 × 8, 4 × 4.
match2_s2	Set 3: GCF of 20 and 30	**10**	Common factors 1, 2, 5, 10. Greatest: 10.
match3_s2	Set 3: LCM of 6 and 8	**24**	Multiples of 6: 6, 12, 18, 24. Multiples of 8: 8, 16, 24. First shared: 24.
match4_s2	Set 3: The only even prime	**2**	Every other even number has 2 as an extra factor.
match5_s2	Set 3: Divisible by 9	**81**	8 + 1 = 9, so 81 is divisible by 9 (81 = 9 x 9). None of the other answers are.

BOSS BATTLE: THE BIG DELIVERY DAY (4 POINTS EACH)

ID	Question	Answer	Why (explain to student)
b1_s0	Set 1: The feed truck comes every 4 days and the vet comes every 6 days. Both came on Day 0. At the same time, an order is due: 24 bags of cheese curds and 36 milk bottles, packed into the greatest number of identical crates with nothing left over. From Day 1 through Day 36, on how many days do the truck AND the vet both come? Number of shared days:	**3**	The LCM of 4 and 6 is 12, so they meet every 12 days: Day 12, Day 24, and Day 36. That is 3 days.
b2_s0	Set 1: The feed truck comes every 4 days and the vet comes every 6 days. Both came on Day 0. At the same time, an order is due: 24 bags of cheese curds and 36 milk bottles, packed into the greatest number of identical crates with nothing left over. Two steps: how many milk bottles go in EACH crate?	**3**	Step 1: the greatest number of crates is the GCF of 24 and 36, which is 12. Step 2: 36 bottles divided by 12 crates = 3 bottles per crate. (2 is the number of curd bags per crate.)
b1_s1	Set 2: This month the feed truck comes every 3 days and the vet comes every 8 days. Both came on Day 0. An order is due: 18 bags of cheese curds and 30 milk bottles, packed into the greatest number of identical crates with nothing left over. From Day 1 through Day 50, on how many days do the truck AND the vet both come? Number of shared days:	**2**	The LCM of 3 and 8 is 24, so they meet every 24 days: Day 24 and Day 48. Day 72 is past Day 50. That is 2 days.

ID	Question	Answer	Why (explain to student)
b2_s1	Set 2: This month the feed truck comes every 3 days and the vet comes every 8 days. Both came on Day 0. An order is due: 18 bags of cheese curds and 30 milk bottles, packed into the greatest number of identical crates with nothing left over. Two steps: how many curd bags go in EACH crate?	**3**	Step 1: the greatest number of crates is the GCF of 18 and 30, which is 6. Step 2: 18 bags divided by 6 crates = 3 bags per crate. (5 is the number of bottles per crate.)
b1_s2	Set 3: This month the feed truck comes every 6 days and the vet comes every 10 days. Both came on Day 0. An order is due: 32 bags of cheese curds and 40 milk bottles, packed into the greatest number of identical crates with nothing left over. After Day 0, on what day do the truck and the vet come together for the SECOND time? Day	**60**	The LCM of 6 and 10 is 30, so they meet every 30 days: first on Day 30, then on Day 60.
b2_s2	Set 3: This month the feed truck comes every 6 days and the vet comes every 10 days. Both came on Day 0. An order is due: 32 bags of cheese curds and 40 milk bottles, packed into the greatest number of identical crates with nothing left over. Two steps: how many milk bottles go in EACH crate?	**5**	Step 1: the greatest number of crates is the GCF of 32 and 40, which is 8. Step 2: 40 bottles divided by 8 crates = 5 bottles per crate. (4 is the number of curd bags per crate.)

Pre & Post sheet answer guide

Many items are open-ended. Grade for genuine understanding and complete sentences, not exact wording.

PRE-LESSON WARM-UP

#	Prompt	Model answer / grading note
1	Multiplication facts that equal 12	1 × 12, 2 × 6, 3 × 4 (also accept the turnarounds 12 × 1, 6 × 2, 4 × 3).
2	First six multiples of 4	4, 8, 12, 16, 20, 24
3	Which number is divisible by 5?	55 (it ends in 5).
4	Factors of 20	1, 2, 4, 5, 10, 20
5	Match the words	factor: a number you multiply to get a product; multiple: the answer when you multiply a number by 1, 2, 3, 4…; product: the answer to a multiplication problem.
6	576 ÷ 24	24 (24 × 24 = 576).

POST-LESSON CHALLENGE

#	Prompt	Model answer / grading note
1	Factor pairs of 42	1 × 42, 2 × 21, 3 × 14, 6 × 7. Any matching stacking plan is correct, e.g., 6 stacks of 7 bales or 7 stacks of 6 bales.
2	Which cow ID is prime?	43. (39 = 3 × 13, 49 = 7 × 7, 57 = 3 × 19.)
3	Trucks every 3 and 4 days	In 12 days. 12 is the least common multiple (LCM) of 3 and 4.
4	16 curd bags and 24 bottles into identical crates	8 crates (GCF of 16 and 24 is 8), each with 2 bags of curds and 3 milk bottles.

#	Prompt	Model answer / grading note
5	Writing: is 87 prime?	He is wrong. Model: "87 is not prime. Its digits add to 15, which is divisible by 3, so 87 is divisible by 3 (87 = 3 × 29). Because it has factors other than 1 and itself, 87 is composite." Look for complete sentences, a correct divisibility rule, and the word composite.

★ CHALLENGE

#	Prompt	Model answer / grading note
1	Every 4, 6, and 9 days	Day 36. Multiples of 9: 9, 18, 27, 36. 36 is the first one that is also a multiple of 4 and 6, so the LCM of 4, 6, and 9 is 36.